KITCHEN SINK

one of everything, with the Markdown that draws it in the README

1.1 TEXT.

Plain Markdown paragraphs justify in the Roman face. **Bold**, *italic*, underline, `typewriter`, H_2O and 10^6 all work inline, and a spaced double hyphen – like this – prints as an en dash. Links keep their words and drop the URL, because paper has no links: see the README.

Inline math sits in the line: the time constant is $\tau = RC$ and the corner is $f_c = 1/(2\pi RC)$.

$$v(t) = V_0\, e^{-t/\tau} \tag{1-1}$$

1.1.1 Lists and code.

- an unordered list item
- a second one, which wraps onto a following line when it runs long enough to need more than one line of the sheet
- a third

1. an ordered list
2. numbers itself

```
V16 N68 E      monitor range, altitude rate, altitude
V57 E          landing radar update permitted
```

1.2 TABLES.

A bare pipe table renders ruled and untitled. Wrap it in a table block to number and title it.

Mode	Gain	Rate
coarse	4.0	2.0 deg/s
fine	0.5	0.2 deg/s

Table 1-1. Thruster Quad Assignments

Quad	Axis	Thrusters	Min impulse
A	pitch, roll	A1U A2D A3A A4F	0.4 N s
B	yaw, roll	B1L B2R B3A B4F	0.4 N s
C	pitch, roll	C1U C2D C3A C4F	0.4 N s
D	yaw, roll	D1L D2R D3A D4F	0.4 N s

1.3 CALL-OUTS.

NOTE

A note is advice: nothing breaks if it is ignored.

CAUTION

A caution protects the equipment. Do not command fine mode with a quad isolated.

WARNING

A warning protects the crew. It is boxed so it cannot be missed.

1.3.1 Change bars.

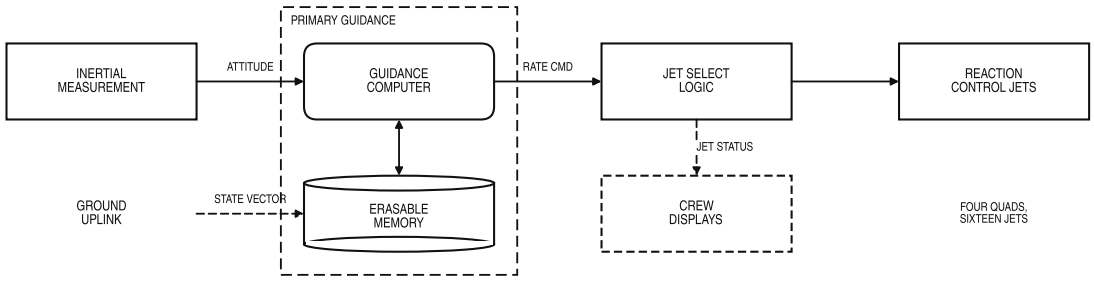
Anything inside a changed block gets a black bar in the left gutter, the handbook's way of saying "this paragraph is new since the last issue".

QUAD ISOLATION	CDR
QUAD A ISOL VLV	CLOSE
verify talkback	BARBERPOLE
RCS SYS A/B	CHECK PRESS
ATT CONTROL	MODE CONT

FIGURES

1.4 BLOCK DIAGRAMS.

Nodes sit on a grid of columns and rows. Shapes: box, round, drum (storage), dash (outside the system) and text (lettering only). A fence draws a dashed boundary round a subsystem; edges may be dashed, two-headed and labelled.

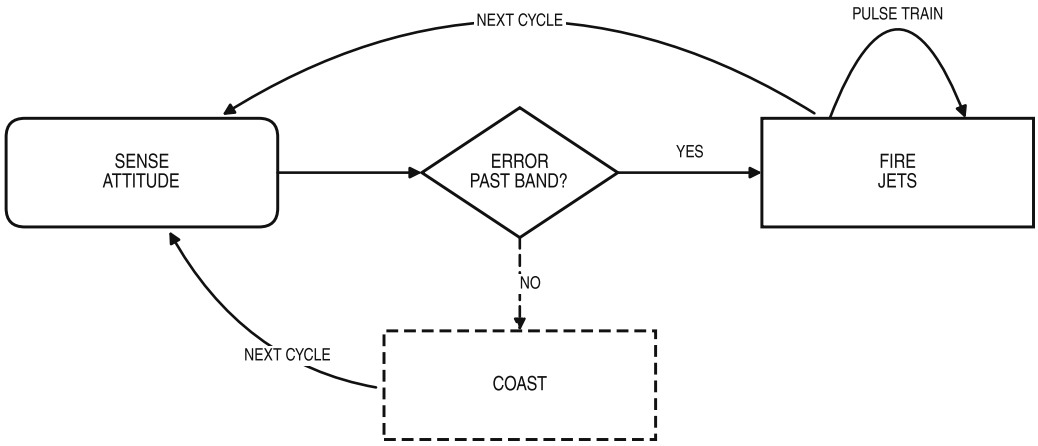


KIT-501A

Figure 1-1. Attitude Control Functional Diagram, Every Shape and Edge

1.5 GRAPHS.

A diagram is also a graph: a diamond is a decision, `bend=` curves a back-edge, and a loop returns a node to itself.



KIT-502A

Figure 1-2. Attitude Hold, a Decision Loop

1.6 TIMELINES.

Bars may be hatched, open or solid; marks are vertical lines with a label above, and a call drops a leader from a bar to lettering below the axis.

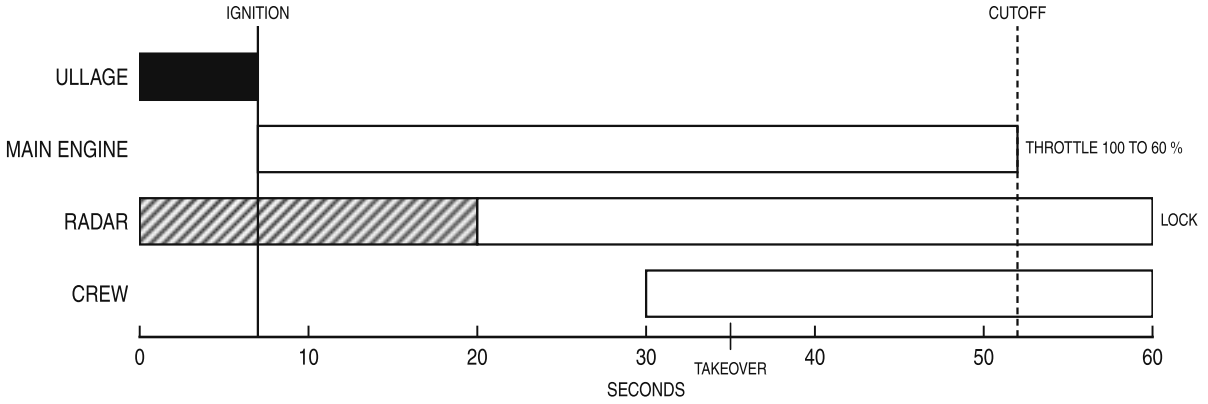


Figure 1-3. Sequence of Events, Every Bar and Mark

1.7 PLOTS.

Lines come from a CSV and are labelled at their right ends instead of with a legend; marks draw a vertical line. Log axes are a single line.

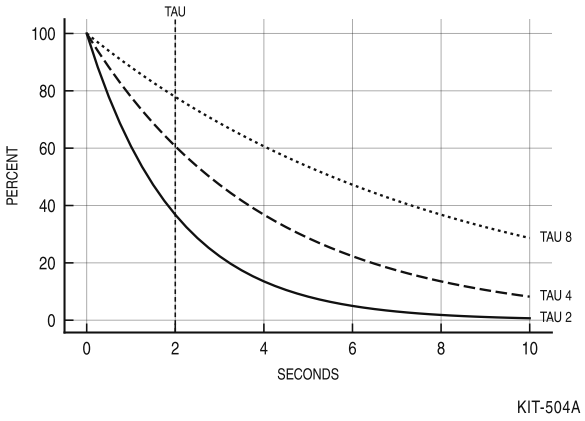


Figure 1-4. Decay of Three Time Constants

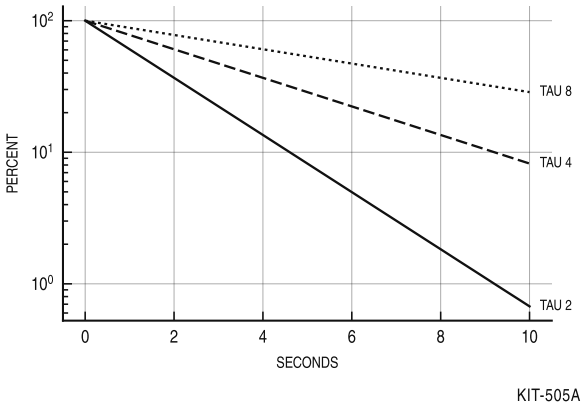


Figure 1-5. The Same, on a Log Axis

1.8 PANELS.

A panel line starts a second panel under the first, on the same x; fn takes degrees() and radians().

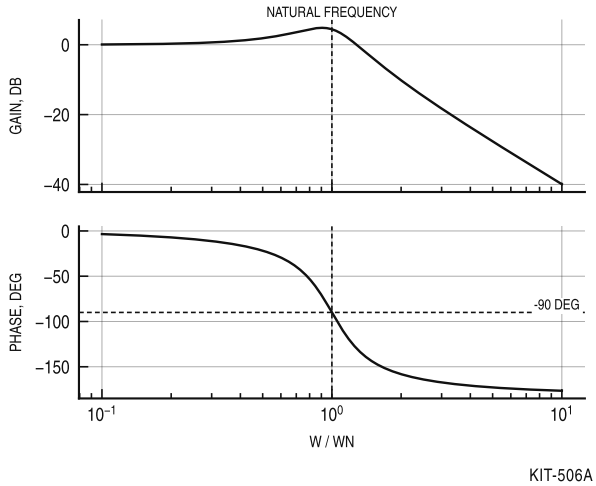
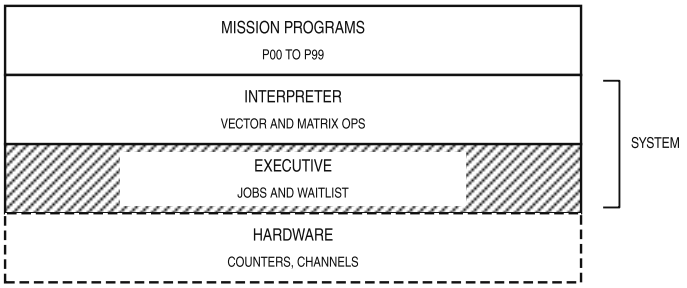


Figure 1-6. Rate Loop, Gain and Phase

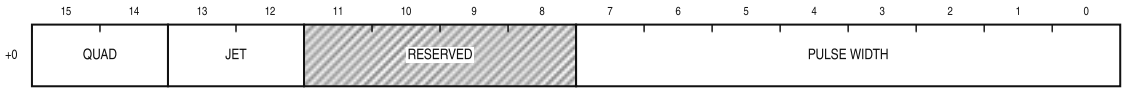
1.9 STACKS AND BIT FIELDS.

A stack sets layers top to bottom; a bit field wraps fields into rows of bits, hatching the reserved ones.



KIT-507A

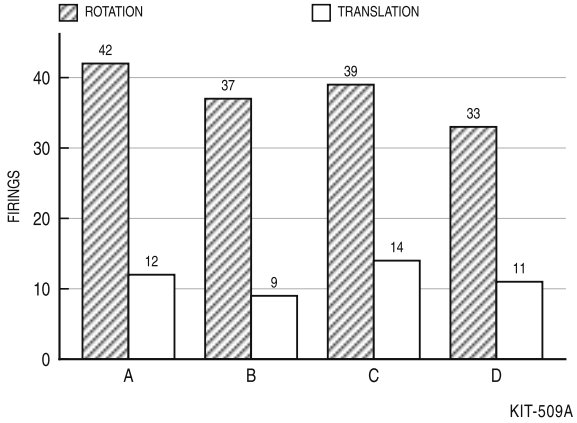
Figure 1-7. Flight Software Layers



KIT-508A

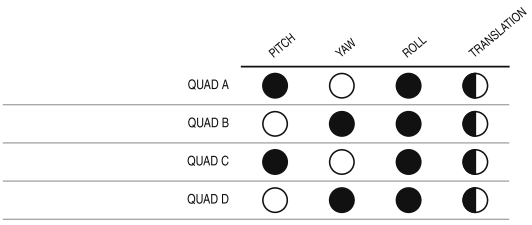
Figure 1-8. Jet Command Word

1.10 BARS AND MATRICES.



KIT-509A

Figure 1-9. Firings by Quad



KIT-510A

Figure 1-10. Quad Authority

With shade, a matrix of numbers is a heatmap: each cell tinted by its value.

	ULLAGE	COAST	BRAKING	APPROACH	LANDING
QUAD A	4	2	9	14	21
QUAD B	4	1	7	12	18
QUAD C	4	2	8	15	20
QUAD D	4	1	6	11	17

KIT-511A

Figure 1-11. Jet Firings by Quad and Phase

1.11 POINTS AND SPREADS.

A scatter letters its points and fits a line; a dist draws boxes from five numbers (or raw CSV values).

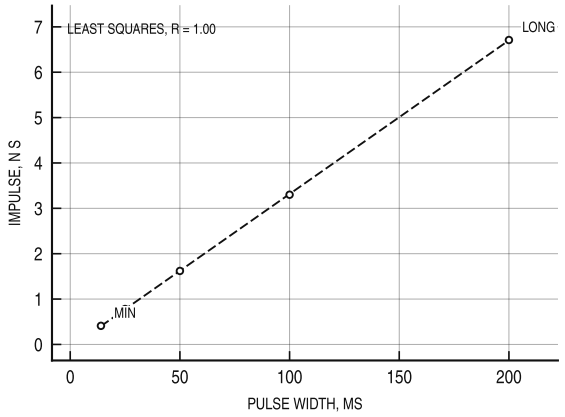


Figure 1-12. Pulse Width and Impulse

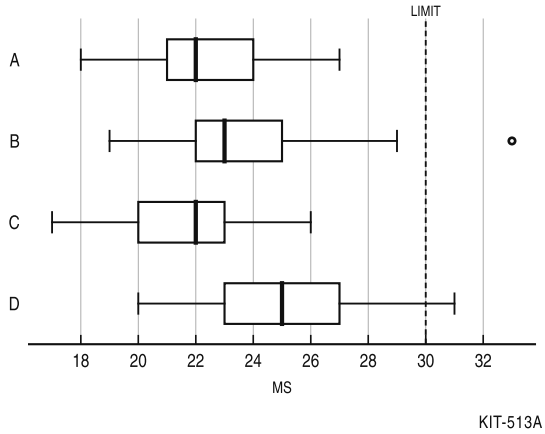


Figure 1-13. Jet Response by Quad

KIT

1.12 FLOWS.

A sankey draws flows between stages as ribbons as wide as their values.

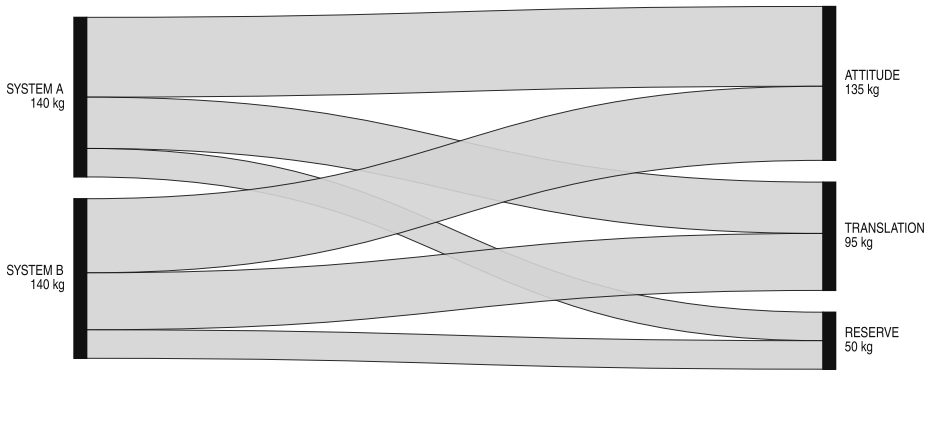


Figure 1-14. Propellant Budget, kg

1.13 IMAGES AND CALLOUTS.

Any SVG or PNG. Width may be half or third; gray=1 prints a dark screen capture as ink on paper. A callouts block draws leaders from points on a picture to lettering in its margins, on the nearer side.

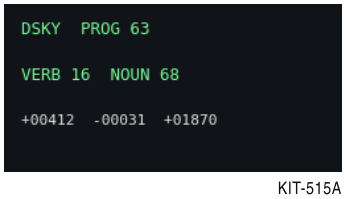


Figure 1-15. A screen capture, as captured

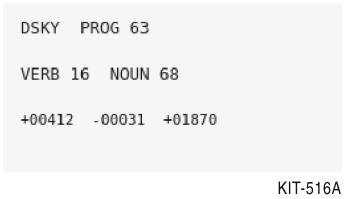
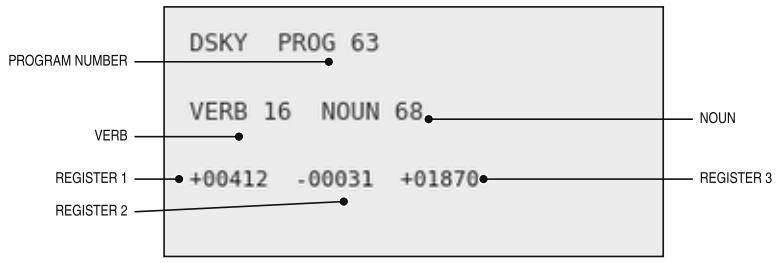


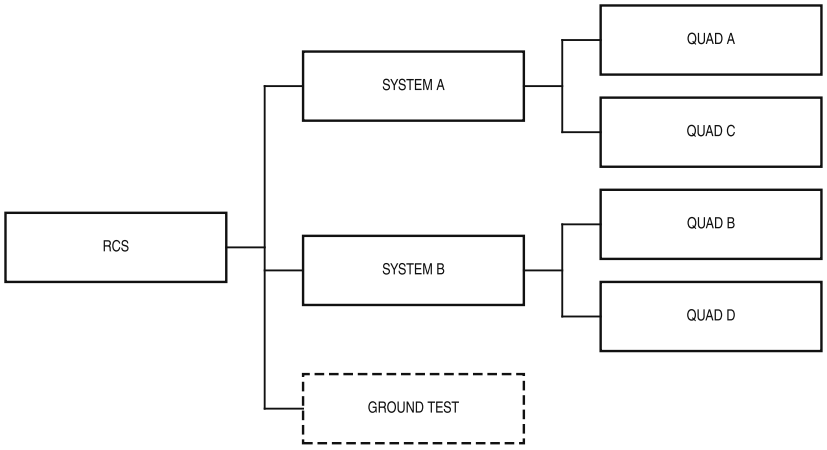
Figure 1-16. The same capture with gray=1



KIT-517A

Figure 1-17. The Display, Named

1.14 TREES.

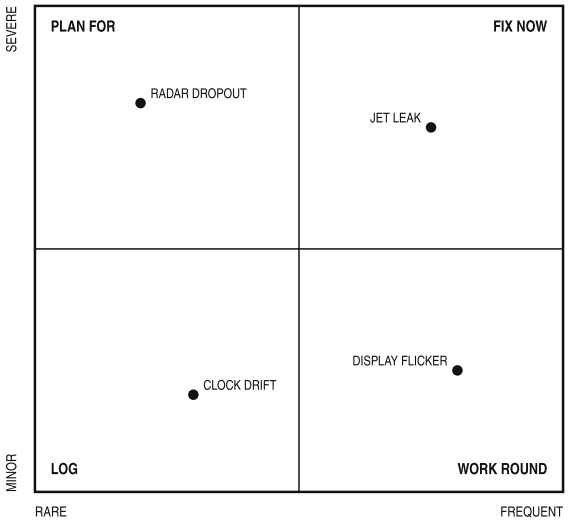


KIT-518A

Figure 1-18. Reaction Control, by System

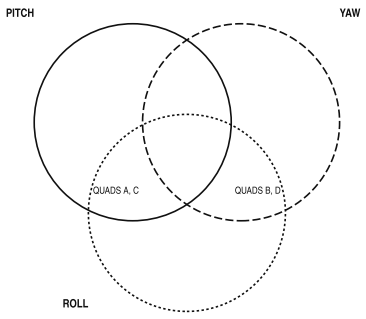
1.15 SHARES AND CHOICES.

A quadrant places items on two judgements; a Venn letters overlaps; a treemap nests shares; a funnel narrows.



KIT-519A

Figure 1-19. Anomalies to Work



KIT-520A

Figure 1-20. Jets by Duty

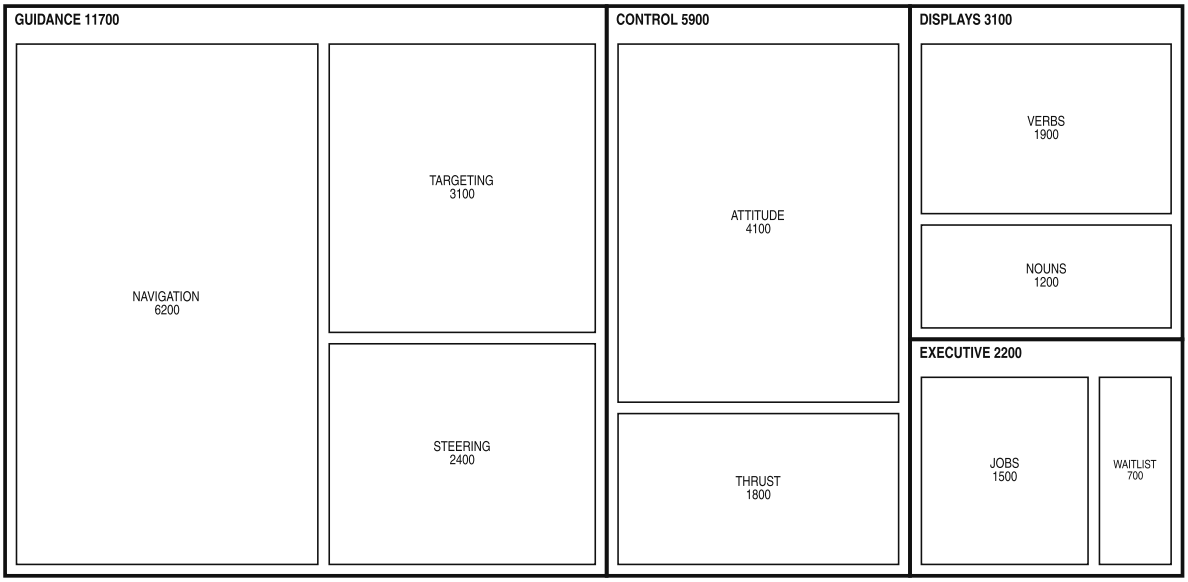


Figure 1-21. Flight Software by Module, Words

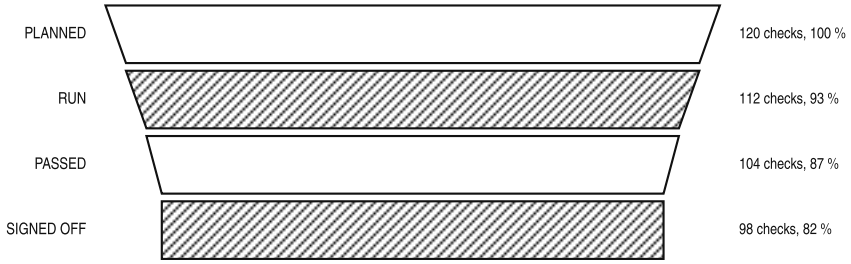


Figure 1-22. Pre-Flight Checks

1.16 TOTALS AND CHANGES.

A waterfall walks a running total; a slope runs each thing from one build to the next; a waffle counts shares.

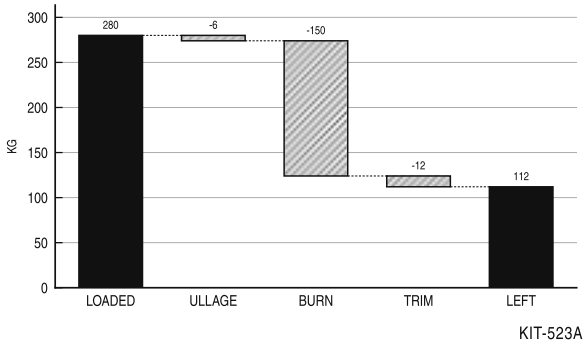


Figure 1-23. Propellant over a Burn, kg

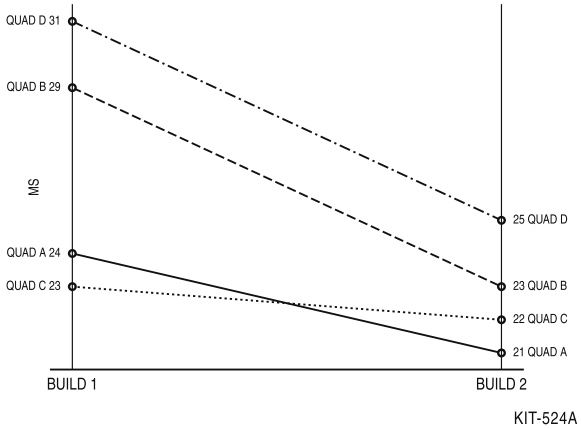


Figure 1-24. Jet Response by Build, ms

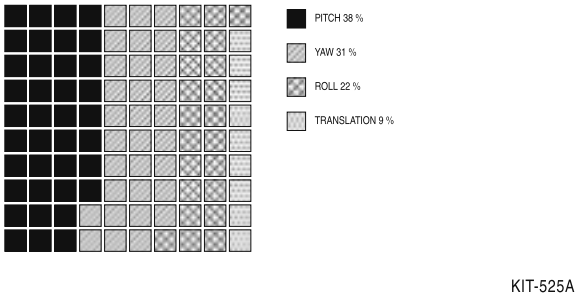


Figure 1-25. Thruster Time by Axis

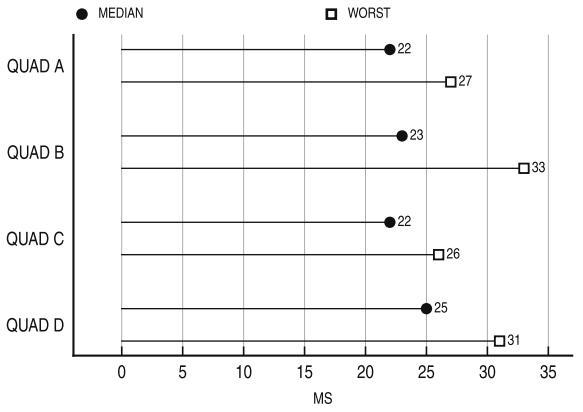


Figure 1-26. Jet Response, Median and Worst

KIT

1.17 FIELDS, DIRECTIONS AND MANY MEASURES.

A contour draws a field with its arrows; a raster sets it as gray; a polar plot draws a pattern or a wind rose; parallel axes give each measure its own axis.

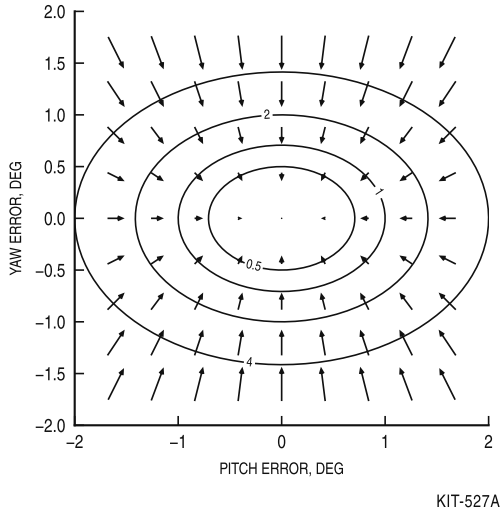


Figure 1-27. Attitude Error Potential

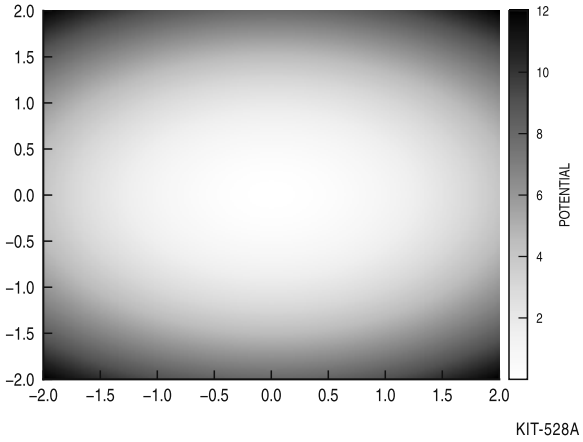


Figure 1-28. The Same, as Gray

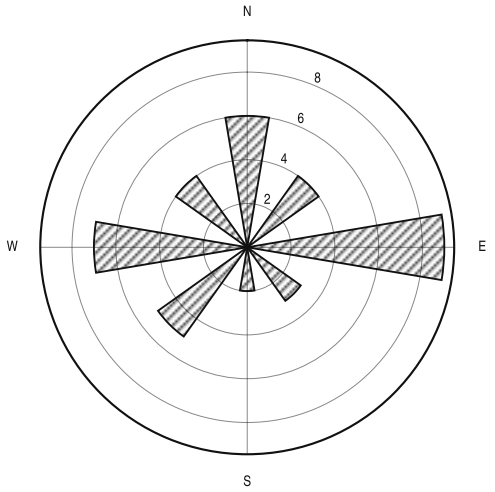


Figure 1-29. Landing-Site Winds, Example

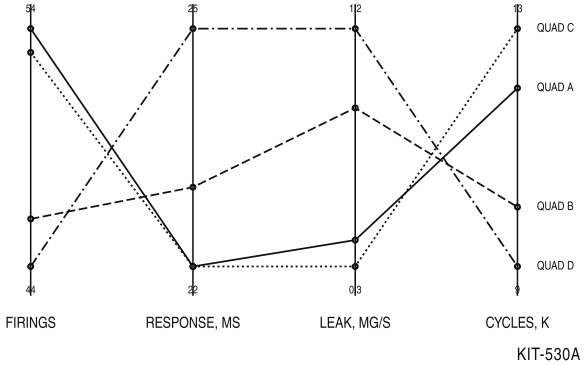


Figure 1-30. Four Quads, Four Measures

KIT

1.18 NETWORKS, LANES AND PLANS.

A network lays out a graph for you; lanes give each row of a diagram an owner; milestones and dependencies make a timeline a plan.

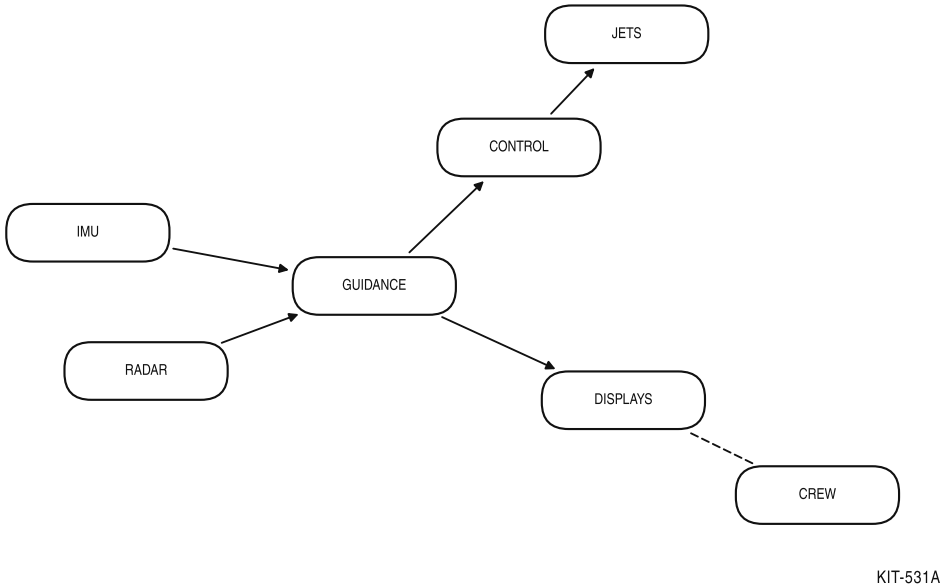


Figure 1-31. Who Talks to Whom

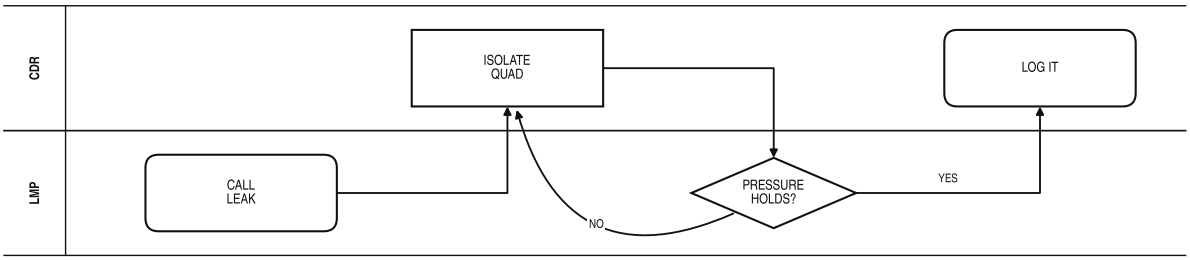


Figure 1-32. Quad Isolation, by Crew Member

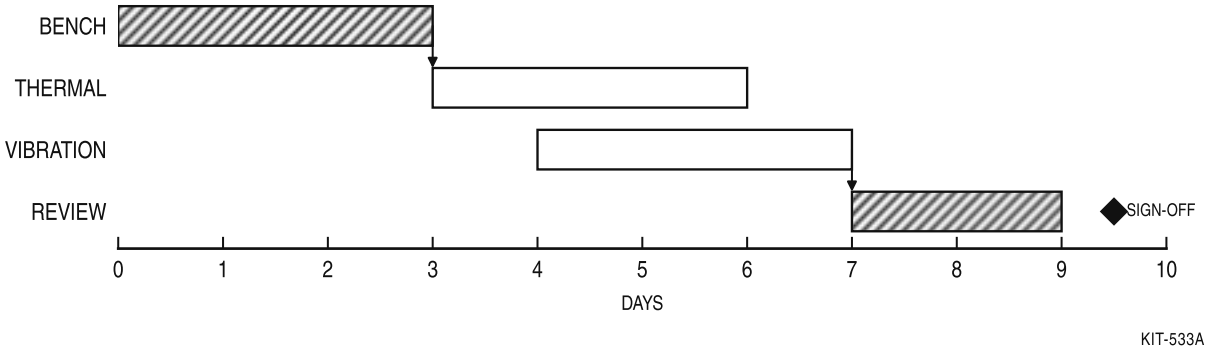


Figure 1-33. Bring-Up Plan, Days

1.19 MAPS AND SPARKLINES.

A map draws GeoJSON or shapefile layers with points, routes, a graticule and a scale. A sparkline sits in a line of text, the last value dotted: quad B's response over six builds ran , ms.

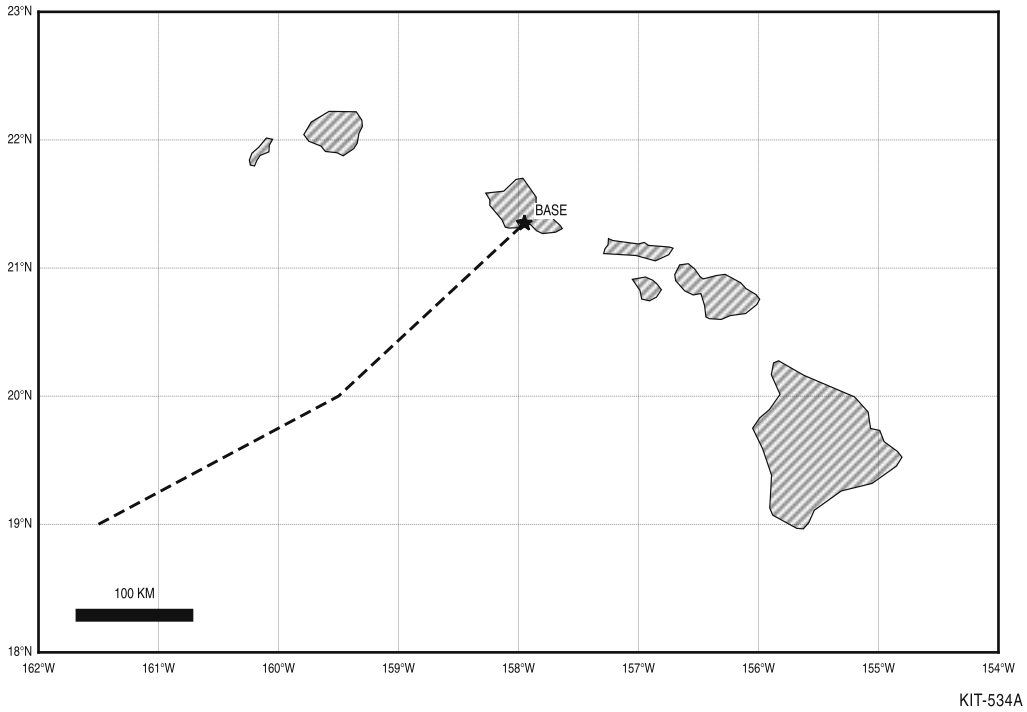


Figure 1-34. Recovery Area, Example